n_TOF X17 Trigger — Timing & Scaler Rates

2026-07-11 (parasitic/dedicated beam)

Configuration as tested

M1 (.240) is offline, so the 20 ns sector-AND coincidence window is imposed at **M3 (.242) input Gate&Delay** on *both* legs (wall = in-ch0, scint = in-ch1), rather than at the M1/M2 output monostables. Trigger chain (verified on the logic analyzer): Singles (M4.A) + Doubles (M4.B) → M4.C OR (no veto) → M4.D OR → DREAM external trigger.

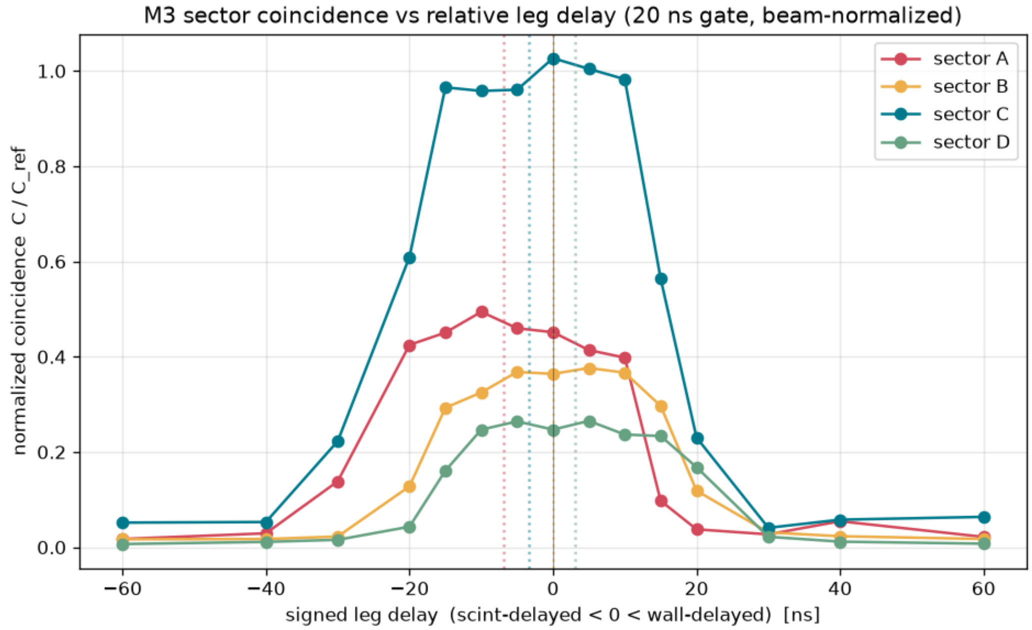
1. Timing scan — do we need delays? What gate width?

Short answer: no per-sector delays; gate width = 20 ns on both M3 input legs.

The delay-curve scan swept the relative wall–scint delay (±60 ns) and measured each sector's coincidence rate, beam-normalized against two held reference sectors (C/C_{ref}). The plateau *center* is the residual skew; all four are within ±7 ns, i.e. $|\text{center}| \leq 10$ ns, so **no delay is applied** — the symmetric 20 ns window is well-centered on every sector. FWHM $\approx$ 36–39 ns $\approx w_1 + w_2$ (two 20 ns pulses), confirming the gate does what we intend. We stayed at 20 ns (did *not* drop to 15 ns, which would clip sector A).

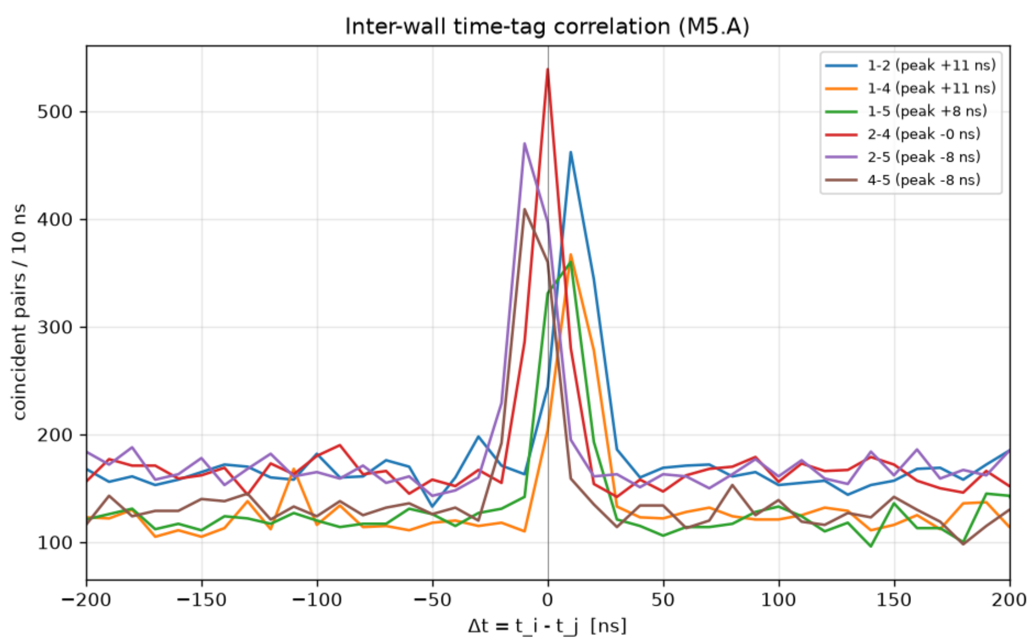
Sector	Plateau center (ns)	FWHM (ns)
A	−6.8	39
B	−0.1	36
C	−3.4	38
D	+3.1	39

Resulting gate/mono settings: M3 input G&D gate = **20 ns**, delay = 0 (both legs, all sectors); M3 output monostables = **30 ns**; M4.B Doubles coincidence window = **50 ns**.



2. Inter-wall timing (Module 5 time-tag)

Every wall pair shows a sharp coincidence peak within ±11 ns of zero (relative to panel 1: panel 2 +11, panel 4 +11, panel 5 +8 ns), all inside the ±20 ns tolerance ⇒ the four walls are mutually aligned; **no M4.A/B trim needed**.



3. Scaler rates (120 s, beam on ~410–850×10^{10} protons)

These are beam-driven and scale with intensity (follow-up pending).

M5 scaler	ch0 (A)	ch1 (B)	ch2 (C)	ch3 (D)	
Walls (M5.A)	601	833	616	669	Hz
Scints (M5.B)	67	46	129	25	Hz
Sectors (M5.C)	11.2	8.3	20.0	5.1	Hz

Trigger-logic outputs:

Signal	Rate
Singles (M4.A OR of sectors)	≈ 43.6 Hz
Doubles (M4.B, ≥2 sectors, 50 ns window)	≈ 0.78 Hz
Doubles / Singles	≈ 0.018

The Doubles are essentially all *real* coincidences: tightening the window 100→50 ns left Doubles/Singles unchanged (0.988×). The M4.B per-input counters [TOTAL, CH1, CH2, CH4, CH5] = [0.78, 0.008, 0.74, 0.025, 0.008] Hz show sector B (CH2) opens ~95% of the coincidence windows.

Plots: `n1081b/snapshots/timing_scan_run2.png`, `walls_tt_v1.png`. Data: `timing_scan_run2.json`, `walls_tt_v1.csv`. Final board state: `dump_2026-07-11_timing_final.json`.